

DBMS – SELLER AND INVENTORY MANAGEMENT

StyleNova – Fashion Products E-Commerce Order Management System

1. Database Design

1.1 Seller Table

Attribute	Data Type	Constraint	Description
Seller_ID	INT	Primary Key	Unique identifier for each seller
Seller Name	VARCHAR(100)	NOT NULL	Name of the seller
Business_Name	VARCHAR(100)	NOT NULL	Business name of the seller
Email	VARCHAR(100)	UNIQUE	Seller email address
Phone	VARCHAR(15)	—	Seller contact number
Address	VARCHAR(200)	—	Seller address

1.2 Inventory Table

Attribute	Data Type	Constraint	Description
Inventory_ID	INT	Primary Key	Unique identifier for inventory record
Product_ID	INT	Foreign Key	References Product.Product_ID
Seller_ID	INT	Foreign Key	References SN_SellerDetails.Seller_ID
Quantity Available	INT	NOT NULL	Available stock quantity
Last Updated	DATE	—	Date of last stock update

2. Primary and Foreign Key Relationship

SN_SellerDetails → SN_InventoryDetails

- SN_SellerDetails.Seller_ID → Primary Key
- SN_InventoryDetails.Inventory_ID → Primary Key
- SN_InventoryDetails.Seller_ID → Foreign Key referencing SN_SellerDetails.Seller_ID
- Relationship: One-to-Many (1:M)
- One seller can manage inventory records for multiple products.

Product → SN_InventoryDetails

- Product.Product_ID → Primary Key

- SN_InventoryDetails.Product_ID → Foreign Key referencing Product.Product_ID
- Relationship: One-to-One (1:1)
- Each product has one inventory record for stock tracking.

Relationship Structure: SN_SellerDetails → Product → SN_InventoryDetails

3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE SN_SellerDetails (  
2     Seller_ID INT PRIMARY KEY,  
3     Seller_Name VARCHAR(100) NOT NULL,  
4     Business_Name VARCHAR(100) NOT NULL,  
5     Email VARCHAR(100) UNIQUE,  
6     Phone VARCHAR(15),  
7     Address VARCHAR(200)  
8 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE SN_InventoryDetails (  
2     Inventory_ID INT PRIMARY KEY,  
3     Product_ID INT NOT NULL,  
4     Seller_ID INT NOT NULL,  
5     Quantity_Available INT NOT NULL,  
6     Last_Updated DATE,  
7     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),  
8     FOREIGN KEY (Seller_ID) REFERENCES SN_SellerDetails(Seller_ID)  
9 );
```

Table created.

4. Data Insertion

4.1 Seller Records

```
SQL> INSERT INTO SN_SellerDetails
  2 (Seller_ID, Seller_Name, Business_Name, Email, Phone, Address)
  3 VALUES
  4 (1, 'Arun Kumar', 'Arun Fashion Store', 'arun@gmail.com', '9876543210', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO SN_SellerDetails
  2 (Seller_ID, Seller_Name, Business_Name, Email, Phone, Address)
  3 VALUES
  4 (2, 'Priya Devi', 'Priya Fashion Hub', 'priya@gmail.com', '9876543211', 'Coimbatore');

1 row created.

SQL>
SQL> INSERT INTO SN_SellerDetails
  2 (Seller_ID, Seller_Name, Business_Name, Email, Phone, Address)
  3 VALUES
  4 (3, 'Karthik Raj', 'Karthik Trends', 'karthik@gmail.com', '9876543212', 'Madurai');

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

4.2 Inventory Records

```
SQL> INSERT INTO SN_InventoryDetails
  2 (Inventory_ID, Product_ID, Seller_ID, Quantity_Available, Last_Updated)
  3 VALUES
  4 (4, 104, 3, 12, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO SN_InventoryDetails
  2 (Inventory_ID, Product_ID, Seller_ID, Quantity_Available, Last_Updated)
  3 VALUES
  4 (5, 105, 2, 10, SYSDATE);
INSERT INTO SN_InventoryDetails
*
```

5. Maintain Seller Product Information

5.1 Display Seller Records

```
SQL> SELECT
2  SELLER_ID,
3  SELLER_NAME,
4  BUSINESS_NAME,
5  EMAIL,
6  PHONE,
7  ADDRESS
8  FROM SN_SellerDetails;
```

SELLER_ID	SELLER_NAME	BUSINESS_NAME	EMAIL	PHONE	ADDRESS
1	Arun Kumar	Arun Fashion Store	arun@gmail.com	9876543210	Chennai
2	Priya Devi	Priya Fashion Hub	priya@gmail.com	9876543211	Coimbatore
3	Karthik Raj	Karthik Trends	karthik@gmail.com	9876543212	Madurai

5.2 Display Inventory Records

```
SQL> SELECT * FROM SN_InventoryDetails;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	QUANTITY_AVAILABLE	LAST_UPDA
2	102	1	15	17-SEP-26
3	103	2	20	17-SEP-26
4	104	3	12	17-SEP-26

5.3 Display Seller, Product and Stock Information

```
SQL> SELECT
2  s.Seller_ID,
3  s.Seller_Name,
4  s.Business_Name,
5  p.Product_ID,
6  p.Product_Name,
7  p.Price,
8  i.Quantity_Available
9  FROM SN_SellerDetails s
10 JOIN SN_InventoryDetails i
11 ON s.Seller_ID = i.Seller_ID
12 JOIN Product p
13 ON i.Product_ID = p.Product_ID;
```

SELLER_ID	SELLER_NAME	BUSINESS_NAME	PRODUCT_ID	PRODUCT_NAME	PRICE	QUANTITY_AVAILABLE
1	Arun Kumar	Arun Fashion Store	102	Denim Jeans	1299	15
2	Priya Devi	Priya Fashion Hub	103	Floral Kurti	899	20
3	Karthik Raj	Karthik Trends	104	Kids Hoodie	749	12

6. Track Available and Unavailable Products

6.1 Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity_Available
  5 FROM SN_InventoryDetails i
  6 JOIN Product p
  7 ON i.Product_ID = p.Product_ID
  8 WHERE i.Quantity_Available > 0;
```

PRODUCT_ID	PRODUCT_NAME	QUANTITY_AVAILABLE
102	Denim Jeans	15
103	Floral Kurti	20
104	Kids Hoodie	12

6.2 Unavailable Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity_Available
  5 FROM SN_InventoryDetails i
  6 JOIN Product p
  7 ON i.Product_ID = p.Product_ID
  8 WHERE i.Quantity_Available = 0;
```

no rows selected

6.3 Product Status

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity_Available,
  5     CASE
  6         WHEN i.Quantity_Available > 0 THEN 'Available'
  7         ELSE 'Unavailable'
  8     END AS Stock_Status
  9 FROM SN_InventoryDetails i
 10 JOIN Product p
 11 ON i.Product_ID = p.Product_ID;
```

PRODUCT_ID	PRODUCT_NAME	QUANTITY_AVAILABLE	STOCK_STA
102	Denim Jeans	15	Available
103	Floral Kurti	20	Available
104	Kids Hoodie	12	Available

7. Inventory Update

Update the available stock quantity of Cotton T-Shirt.

```
SQL> -- 3. Update Inventory Stock
SQL> UPDATE SN_InventoryDetails
2 SET Quantity_Available = 30,
3     Last_Updated = SYSDATE
4 WHERE Product_ID = 101;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.

SQL>
SQL>
SQL> -- 4. Verify the updated record
SQL> SELECT
2     Inventory_ID,
3     Product_ID,
4     Seller_ID,
5     Quantity_Available,
6     Last_Updated
7 FROM SN_InventoryDetails
8 WHERE Product_ID = 101;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	QUANTITY_AVAILABLE	LAST_UPDA
1	101	1	30	17-SEP-26

8. Inventory Status Reports

8.1 Complete Inventory Report

```
SQL> SELECT
2     i.Inventory_ID,
3     s.Seller_Name,
4     p.Product_Name,
5     i.Quantity_Available,
6     CASE
7         WHEN i.Quantity_Available > 0 THEN 'Available'
8         ELSE 'Unavailable'
9     END AS Stock_Status,
10    i.Last_Updated
11 FROM SN_InventoryDetails i
12 JOIN SN_SellerDetails s
13 ON i.Seller_ID = s.Seller_ID
14 JOIN Product p
15 ON i.Product_ID = p.Product_ID;
```

INVENTORY_ID	SELLER_NAME	PRODUCT_NAME	QUANTITY_AVAILABLE	STOCK_STATU	LAST_UPDA
2	Arun Kumar	Denim Jeans	15	Available	17-SEP-26
3	Priya Devi	Floral Kurti	20	Available	17-SEP-26
4	Karthik Raj	Kids Hoodie	12	Available	17-SEP-26
1	Arun Kumar	Cotton T-Shirt	30	Available	17-SEP-26

8.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     CASE
  3         WHEN Quantity_Available > 0 THEN 'Available'
  4         ELSE 'Unavailable'
  5     END AS Stock_Status,
  6     COUNT(*) AS Product_Count
  7 FROM SN_InventoryDetails
  8 GROUP BY
  9     CASE
 10         WHEN Quantity_Available > 0 THEN 'Available'
 11         ELSE 'Unavailable'
 12     END;
```

STOCK_STATU	PRODUCT_COUNT
Available	4

8.3 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Quantity_Available) AS Total_Stock
  5 FROM SN_SellerDetails s
  6 JOIN SN_InventoryDetails i
  7 ON s.Seller_ID = i.Seller_ID
  8 GROUP BY
  9     s.Seller_ID,
 10     s.Seller_Name;
```

SELLER_ID	SELLER_NAME	TOTAL_STOCK
1	Arun Kumar	45
2	Priya Devi	20
3	Karthik Raj	12

8.4 Products with Zero Stock

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Quantity_Available
  6 FROM SN_InventoryDetails i
  7 JOIN Product p
  8 ON i.Product_ID = p.Product_ID
  9 JOIN SN_SellerDetails s
 10 ON i.Seller_ID = s.Seller_ID
 11 WHERE i.Quantity_Available = 0;
```

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME
101	Cotton T-Shirt	Arun Kumar